

**CONTROL OF THE PARASITIC MITES *Varroa destructor* AND
Acarapis woodi IN HONEY BEE (*Apis mellifera*) COLONIES USING
BOTANICAL COMPOUNDS AND INTEGRATED PEST
MANAGEMENT**

by

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B.Sc., Northland College, 1998

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
Master of Pest Management

In the Department
of
Biological Sciences

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SIMON FRASER UNIVERSITY

September 2001

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Abstract

Two major mite parasites threaten honey bee populations world-wide, the external mite *Varroa destructor* (Anderson and Trueman) and the internal tracheal mite *Acarapis woodi* (Rennie). *Varroa* are developing resistance to the primary acaracide fluvalinate currently used for control, and the need for alternative management methods is critical. Previous studies have shown that botanical compounds, especially thymol, can be effective although variable management tools. Non-chemical control methods also have some effect on varroa populations when used alone, but the efficacy of these methods when used in combination has not been examined.

In the first series of experiments, six release devices for either thymol or a blend of botanical oils known as Magic 3 were tested in beehives. The release devices were 1) Low Density Polyethylene (LDPE) sleeves filled with Magic 3, 2) Magic 3-infused florist blocks, 3) thymol-infused florist blocks, 4) a canola oil and thymol mixture wick release, 5) a plastic strip coated with calcium carbonate and Magic 3, and 6) an untreated control. There were significant decreases in varroa levels with the use of Magic 3 sleeves placed between brood frames, but brood levels also decreased. Tracheal mite levels significantly declined with the Magic 3 sleeve treatment, the Magic 3 florist block treatment, and the thymol canola wick treatment. Changing location of Magic 3 sleeves in the colony eliminated impact on honey bees, but also did not effectively control varroa mites.

In the second series of experiments, hygienic bee stock, modified bottom boards, and thymol infused florist blocks in a 1) mixed IPM treatment were compared to colonies that received single or dual rotated chemical controls, either 2) thymol in the spring and

thymol in the fall, 3) Apistan in the spring and Apistan in the fall, 4) thymol in the spring and Apistan in the fall, or 5) Apistan in the spring and thymol in the fall.

Adult bee population, brood area, mite population, honey production, and colony loss over the period of the experiment were measured to determine overall colony health and treatment impact. During the 21 months that the IPM experiment was conducted, there were no significant differences in adult bee population, brood area, honey production, or colony loss between any of the treatments. There was a significant difference in varroa mite population only during the post-treatment measurement period in the fall of 2000, with the Apistan/Apistan and thymol/Apistan having significantly lower mite levels than the thymol/thymol and Apistan/thymol treatments. These results suggest that a mixed IPM strategy or a rotational IPM regime could be viable for varroa control.

Dedication

For my parents. I couldn't ask for two more supportive and caring people in my life.

Also. . .

For everyone who helped define my SFU experience. The academic knowledge will (hopefully) last a lifetime, and the personal insights all of you provided are priceless.

Warm fuzzies for all!

As for the thesis. . .

Kind of like sniffing a couple hundred thousand magic markers while getting a slow sheet-lightning enema, isn't it?

-Max Cannon

Acknowledgments

I would like to thank the members of the Simon Fraser University “Swarm Team” for their valuable assistance, especially Shawn Devlin, Heather Higo, Kerri Johannson, Adony Melathopoulos, Hien Ngo, Stephen Pernal, Holly Sabara, and Robin Whittington. I also would like to thank Murray Isman, Rod Bradbury, Eco-Smart and Eco-Safe Technologies for advice, preparation of release devices and donation of Magic 3, and the British Columbia Ministry of Agriculture and Food for the use of an apiary site in Abbotsford, BC. Funding was provided by Eco-Smart Technologies, The Agri-Food Innovation Fund from Saskatchewan Agriculture, the Natural Sciences and Engineering Research Council, the Canadian Bee Research Fund, and the British Columbia Honey Producers Association.

Beyond the confines of the academic realm, I would like to offer my deepest thanks to Heather Higo, Robin Whittington, Shawn Devlin, Holly Sabara, Monique Ledoux, Darnell Hagen, Bob Lewis, Alex Gustafsson, and Cayetana "Tania" Schluter for being such good friends. I don't think I've ever laughed so much in my life. None of this would have been possible without them. On the flip side, very much in the confines of academia, I would like to thank Russell Nicholson for his valuable comments during the formation of this thesis and for the papers I submitted during the course of this project. My work is stronger because of his influence. I'm not sure "thank you" is a strong enough sentiment for my gratitude towards Mark Winston for his unending patience, careful guidance, and thoughtful advice. I'm not the same person leaving as I was when I came here and I think Mark has a lot to do with that. . . also.

And Mis. What can I say that hasn't already been said?

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Chapter I

Introduction

Honey bee colonies are being affected world-wide by parasitic mites, in particular *Varroa destructor* (formerly known as *Varroa jacobsoni* Oudemans, recently reclassified by Anderson and Trueman [2000]) and *Acarapis woodi* (Rennie). Both mites have spread throughout the United States and much of Canada since their introductions to North America in the 1980's (De Jong 1997, Wilson et al. 1997). Their expansion has been facilitated by the mobile nature of commercial beekeeping, including the transport of bees from continent to continent and within countries. *Varroa* initially was a parasite of the Asian honey bee *Apis cerana*, but due to introduction of *Apis mellifera* into Asia and Eastern Russia, the two species came into direct contact with each other. By 1960 varroa was found in Asian *A. mellifera* colonies, and subsequently spread across Europe and North America (De Jong 1997).

Varroa infests a colony and establishes itself quickly. The mites are phoretic and feed on adult bee hemolymph until they find a brood cell with a larva that is within 20 to 40 h of being sealed (De Jong 1997). The female mite will lay five to six eggs in the cell, the first of which is male, the rest female. The mites hatch and undergo two nymphal stages after which they become reproductively viable. The mated females emerge with the bee, but the male and any females still in a nymphal stage die. Mites preferentially choose to infest the larger drone larvae cells over worker larvae cells, and reproductive success is higher for varroa that infest drone larvae (De Jong 1997, Martin 1997).

Feeding by the mites results in malnourished larvae, greater larval susceptibility to viral infections that weaken and deform emerging bees, and a decrease in the adult bees' ability to perform tasks. Eventually colonies no longer have sufficient capacity to gather resources, rear brood, and survive (Martin 1997, Donze et al. 1998). Pollination capabilities are lowered and crop production suffers when colonies are disabled, causing severe economic loss to farmers as well as beekeepers (Delaplane and Mayer 2000).

Transfer of mites from colony to colony occurs when a bee hosting a phoretic mite comes into contact with another bee during robbing, drifting, or swarming. Spread of mites throughout an apiary can occur rapidly, and feral colonies established by swarms succumb quickly to mite infestation, which has severely reduced feral honey bee populations in North America.

Registered treatments to control varroa in managed colonies include fluvalinate, amitraz, (used in Europe), coumaphos, and more recently formic acid, which is registered as a liquid in Canada and as a gel formulation in the U.S. (Feldlaufer et al. 1997, Kochansky and Shimanuki 1999). Honey bee colonies in North America are most commonly treated for varroa infestations with PVC strips (Apistan®) inundated with a 10% solution of the synthetic pyrethroid fluvalinate, which can result in up to 99% mite mortality. However, resistance to this pesticide appeared in Europe in 1992 (Eischen 1995), in Florida and South Dakota in 1997 (Elzen et al. 1998), and now can be found in several areas throughout the world (Faucon et al. 1995, Hillesheim et al. 1996, Londzin and Sledzinski 1996, Colin et al. 1997, Trouiller 1998, Mozes-Koch et al. 2000). Varroa control with fluvalinate has

dropped from the 95 - 99% level to as little as 5.1% in some regions (Elzen et al. 1998).

The organophosphorous compound coumaphos (Check-Mite®) is now used in the U.S. under emergency license for mite control in areas where resistance to Apistan is high (Sanford et al. 1999). However, mites are beginning to develop increased tolerances to coumaphos and other acaracidal substances (Milani 1999, Elzen et al. 2000), so its utility may be short-lived.

The use of chemicals in the colony also could tarnish the "natural" reputation of hive products. Fluvalinate residues have been found in both wax and honey, but since this chemical is lipid-soluble, residues are mostly a problem in wax (Balayannis and Santas 1992, Wallner 1995, Bogdanov et al. 1998b). Coumaphos is also lipid-soluble and leaves residues in wax but not in honey at a level that is considered problematic for human consumption (Thrasyvoulou and Pappas 1988, Wallner 1995, Bogdanov, et al. 1998b).

Honey bees also are parasitized by the internal tracheal mite *Acarapis woodi* (Rennie). This mite, described from the Isle of Wight, UK in 1921, was thought to be responsible for what was then known as "Isle of Wight disease," first noted in 1905. Isle of Wight disease caused bee deformity, bee paralysis, and colony mortality (Wilson et al. 1997), symptoms now understood to be caused by tracheal mites. The mite became established throughout most of Europe between 1920 and 1930 (Kjer et al. 1989) and was discovered in Mexico and the United States in 1980 and 1984 respectively (Delfinado-Baker 1984). The tracheal mite is an internal parasite and dies within a few hours after being separated from its host. Except for transfer from

one bee to another, tracheal mites undergo their entire life cycle within the trachea of the bee and the feeding of mites causes scarring and hardening of the tracheal walls.

Tracheal mites prefer younger bees and will transfer from older infested bees onto younger uninfested bees during normal bee-bee contact within the hive. An older bee infested with tracheal mites may come into contact with an uninfested bee from another colony and mites may transfer, generally during robbing, drifting of individual workers and drones, and swarming (Wilson et al. 1997).

Menthol, formic acid, and vegetable oil patties can control tracheal mites (Herbert et al. 1987, Smith et al. 1991, Delaplane 1992, Sammataro et al. 1994). Menthol crystals are placed in a small pouch and the vapors produced are breathed in by the bees, killing the mites inside the trachea. Formic acid is applied using a soaked absorbent pad, although gel formulations to minimize applicator hazard are becoming available. Like menthol, the fumes from the acid are inhaled by the bees, killing the mites in the trachea. Vegetable oil patties are used to disrupt female mites' ability to disperse to young bees. Both menthol and formic acid are not well-dispersed at lower temperatures, and both must be applied more than once. Menthol kills only the adult mites and has no effect on the eggs and nymphal stages in the trachea (Wilson et al. 1997).

Selection can quickly yield honey bee stock that is resistant to tracheal mites (De Guzman et al. 1998, Sherman et al. 1998), providing a non-chemical control option for many beekeepers. The mechanism of bee resistance to tracheal mites is not known, but it is suspected that grooming behavior, which also confers some resistance to varroa, may lower tracheal mite levels. Tracheal mite infestation is

chronic. Once a colony is infested low levels of mites will persist, and eradication is not possible with current methods.

Tracheal mites decrease adult bee longevity, which is correlated with low honey production and increased winter mortality. The normal five to ten percent winter colony loss experienced in apiaries can increase to 50-100% loss with tracheal mites (Wilson et al. 1997). Colony loss is significantly higher with simultaneous infestations of varroa and tracheal mites than with infestations by either mite alone (Downey et al. 2000). If the tracheal mite does not kill the colony directly, it weakens the bee population so that varroa impact increases.

Both varroa and tracheal mites have affected how bees are kept and in some cases have made beekeeping non-viable for hobbyists and commercial operations. Beekeepers in the U.S. and Canada have been facing an economic crisis caused by low honey prices, and have difficulty recovering the costs for mite treatments (Foreign Agricultural Services branch of the United States Department of Agriculture 2001; Dave Pearen, Agriculture and Agri-Food Canada pers. comm. 2001). In Canada, honey costs have been recovering, but production of statistical data on honey production in the U.S. was discontinued in 2000. Pollination is affected not only due to a decrease in the availability of managed honey bee colonies, but also because feral honey bee colonies supplement the pollination efforts of managed honey bee colonies. However, varroa has decimated feral populations (Delaplane and Mayer 2000). Queen production also has been affected because beekeeping operations in the southeastern U.S. that shipped queens and packaged bees across North America have been forced out of business due to decreased demand for bees from mite-infested areas (Wilson et al. 1997).

Many locations that were considered mite-free, such as New Zealand, now have mite infestations in their colonies and quarantines are in place limiting the movement of honey bees between different geographical areas (Anonymous 2000). The “free roaming” beekeeping of the past has become less of an option as these pests continue their spread, and many beekeepers are limiting apiaries to their specific geographic locations. For example, beekeepers from British Columbia are purchasing locally produced queens instead of importing queens from off shore. Given the uncertain economic status of the industry and the increasingly negative public reaction to pesticide use, the addition of chemical residues at *any* level could reduce demand for bee products to a point where beekeeping for honey production is no longer an economically viable venture.

Beekeepers need a mite control method that is effective, safe, consumer friendly, cost-effective, and relatively easy to apply and use. The objectives of these experiments were to develop and test various release devices for botanical controls to determine their efficacy at controlling varroa and tracheal mites, and then to take the most effective of these devices and test it in an Integrated Pest Management (IPM) treatment that would combine cultural and natural-product control methods.

Chapter II

Botanical Compounds and Mite Control

Co-evolution between insects and plants is a common occurrence. Plants synthesize chemicals naturally for defense; insects develop mechanisms to overcome this protection, and plants further increase their defensive capabilities. This process has been exploited by humans for insect control, and some chemicals that are commonly used today for pest control have been synthesized from botanical compounds. For example, fluvalinate, the active ingredient of Apistan, is a synthetic pyrethroid, and pyrethrins were originally isolated from chrysanthemums. Other botanical compounds are being investigated for a variety of pest control purposes including mite control in honey bee colonies.

Recent experiments examined the potential of botanical compounds as varroa control agents. Melathopoulos et al. (2000a, 2000b) and Naumann and Isman (1996) explored using neem oil as a miticide in bee colonies, and many other researchers have examined a number of different botanical compounds and application methods (Marchetti et al. 1984, Colin 1990, Calderone et al. 1991, Smith et al. 1991, Delaplane 1992, Imdorf et al. 1994, Kraus et al. 1994, Sammataro et al. 1994, Eischen 1996, Imdorf et al. 1996, Calderone et al. 1997, Lindberg et al. 2000).

Laboratory screening assays have been performed on hundreds of different botanical compounds to determine which compounds would potentially be useful as miticide agents (Kraus et al. 1994, Lindberg et al. 2000). Their "efficacy window" is the dose that will kill mites but not have a negative effect on either adult bees or brood. For example, Apistan has a large efficacy window in that the dose required to kill mites is

1000 times lower than the dose that will have a negative effect on bees. For many botanical compounds the dose required to kill mites is higher than the dose that kills bees, but others such as neem and thymol have a useful window.

Experiments involving botanical compounds have shown variable success in both varroa and tracheal mite control, using diverse application methods. For tracheal mite control, menthol has been used effectively when applied as crystals in a small pouch and allowed to volatilize in the colony (Wilson et al. 1988, Delaplane 1992), and vegetable oil soaked filter paper has shown some effect on limiting the ability of the tracheal mite to disperse to new bees (Smith et al. 1991).

Thymol has emerged as the botanical compound that has the largest efficacy window and also provides some control of varroa mites with the least impact on bees. Thymol has been applied as a powder, finely divided crystals much like powdered sugar, sprinkled over the top bars of a colony or suspended in a small pouch within the colony (Marchetti et al. 1984, Chiesa 1991), a fumigant (Colin 1990, Ellis and Baxendale 1997), mixed with olive oil and soaked on cardboard strips (Sammataro et al. 1998), on vermiculite florist blocks (Calderone et al. 1997, Whittington et al. 2000), on vermiculite tablets in a product called Apilife-VAR (Imdorf et al. 1994, Imdorf et al. 1995, Imdorf et al. 1996), on a hardened sponge cloth in a product called Thymovar (Andermatt 1999), and as a gel in a product called Apiguard (Mattila and Otis 1999, Mattila et al. 2000), none of which are registered for use in North America.

With the exploration of different application methods came other tests such as residue analysis and toxicity tests. Thymol leaves residues in wax and honey (Lodesani et al. 1992, Bogdanov et al. 1998a) so the same precautions used with other chemicals

such as not applying treatments during honey flow must also be used with botanicals.

Thymol has been tested for possible mutagenic and toxic effects in order to determine its safety for human use and has been determined to be a safe compound (Karpouhtsis et al. 1998). The main concern with the use of botanicals such as thymol is that they may alter the flavor of honey if present at high concentrations.

Thymol is a monoterpene, and these substances are the main components of many essential oils that are produced by plants to provide toxic, feeding deterrent, or oviposition deterrent effects on insects attempting to exploit the plant for its resources (Cornelius et al. 1997). Some insects also produce monoterpenes that function as pheromones, but this is less common (Ellis and Baxendale 1997, Cornelius et al. 1997). Most monoterpenes have low mammalian toxicity which makes them a good potential resource for insecticide testing (Cornelius et al. 1997).

Both thymol and Magic 3, a proprietary blend of five botanical compounds developed by Eco-Safe Technologies (Saanichton, B.C., Canada) and Murray Isman (University of British Columbia, Vancouver, B.C., Canada), exhibit high varroa mortality levels coupled with low bee mortality in laboratory tests (Lindberg et al. 2000). Therefore, these two substances were selected for further experiments to determine their effects on both parasitic mites. Canola oil also provides tracheal mite control through dispersal disruption and is a good carrier for thymol (Smith et al. 1991). I compared the efficacy of these three oil blends in various delivery devices and locations in the colony with the objective of including the best compound in an IPM system for mite control.

Materials and Methods

Assessments. For both experiments, the adult and brood populations of all test colonies were assessed the week before the treatments commenced, following the protocol outlined in Burgett and Burikam (1985). The assessment procedure was identical for both experiments. After the 24 d treatment period, adult and brood populations were again assessed and all colonies were treated with Apistan (Wellmark International, Dallas, TX) strips for 6 weeks, following label directions. Varroa mites were collected using thin cardboard sheets coated with a non-drying latex glue, and inserted onto the bottom board of the colony. These sheets (sticky boards) were overlaid with a wooden frame covered with wire mesh that allowed mites to fall through but prevented bees from contacting the sticky-board. Sticky boards were placed on the bottom boards of all colonies to capture mites, and removed and replaced once a week for 4 weeks or until mite counts on a board were consistently less than ten, at which point mite measurements were stopped. At the end of the 6 week Apistan application, strips were removed from the colonies.

Data Analysis. Mite drop due to treatments was defined as the number of mites collected on sticky boards during treatment exposure divided by the total number of mites collected during the entire experiment (mites killed during treatment plus mites killed during Apistan application). Brood level change was calculated as the increase or decrease of post treatment brood levels compared with pre-treatment brood levels. Similarly, adult bee population change was calculated as the increase or decrease of post treatment adult bee population as compared with pre-treatment adult bee levels. Unless

otherwise stated, means of all data were compared using the Tukey-Kramer HSD multiple comparison test ($\alpha = 0.05$) (JMP IN, SAS Institute Inc. 1997). Proportion data were log transformed.

Experiment 1.

Colonies. 48 colonies were placed in an apiary located in Abbotsford, B.C. The experimental treatment period began on 22 April 1999 and ended 16 May 1999. Colonies were created by splitting from existing colonies, and each was founded with a newly mated queen imported from Australia. All experimental colonies were kept in one standard Langstroth deep hive body and had three frames each of brood and adult bees, two frames of honey, one feeder frame, and three empty frames. Varroa levels in these 48 colonies were assessed using a 24-h natural drop; that is a test in which no acaricides are used, and mites that fall naturally off the bees are captured on sticky boards and counted. To equalize varroa mite levels, the 18 colonies with the highest mite levels and the 18 colonies with the lowest mite levels had 200g of bees taken from the most highly infested colony and placed in the least infested colony, and vice versa continuing up the rank: second highest with second lowest, third highest with third lowest, etc. Colonies with middle levels of varroa were not manipulated.

Mite Sampling. Sticky boards were removed and replaced once every 4 days, and the varroa mite numbers counted. Tracheal mites were sampled by collecting approximately 30 bees in ethanol from the inner cover of each colony, immediately before the treatments were applied and shortly after they were removed for the final time.

Bee dissections were carried out as outlined by Shimanuki and Knox (1991), and mites were scored as either present or absent in the trachea.

Treatments. The six treatments examined were: 1) Low Density Polyethylene (LDPE) sleeves (Pherotech, Delta, B.C., Canada) filled with Magic 3 prepared by Rod Bradbury (Eco-Safe Technologies, Saanichton, B.C., Canada); 2) plastic strips coated with calcium carbonate and Magic 3; 3) Magic 3 infused florist blocks; 4) thymol (Sigma-Aldrich Canada, Oakville, ON.) infused florist blocks (considered a positive control); 5) canola oil and thymol mixture wick release; and 6) untreated control. Eight colonies per treatment were used, and treatments were assigned randomly. The release devices were replaced at 4-day intervals for 24 days.

Before treatment formulations were selected for experimentation, release rate data was collected in order to determine whether a specific formulation would provide a consistent release rate, and for how long the formulation would maintain that release rate. Some formulations, such as thymol in a LDPE sleeve, did not have a release rate that warranted further experimentation. This is the reason that these six treatments were chosen for this experiment.

Low Density Polyethylene (LDPE) Sleeve. Doses of Magic 3 (16.4 ml) were poured onto strips of cotton batting placed in LDPE sleeves to prevent pooling in the bottom. These sleeves had a release rate of 0.2 ml of Magic 3 per day (unpublished data). Therefore, three sleeves were used to obtain the desired release rate of 0.6 ml per day. These sleeves were hung between the brood frames with a piece of wire looped through a hole in the sleeve. The hole was not in a section of the sleeve containing the treatment.

The sleeves were placed in the center of the brood rearing frames so that they contacted as much of the brood as possible (Fig. 1A).

Calcium Carbonate Paste. Eco-Safe Technologies (Saanichton, B.C., Canada) provided a mixture of 10% Magic 3, 10% Linseed oil, and 80% calcium carbonate coated on a 15 cm x 2 cm x 2 mm thick strip. Each strip had approximately 2 mm of paste on each side, which gave an approximate dosage of 1.3 ml of Magic 3 per strip. These strips had a release rate of 0.05 ml per day. Three strips per colony were hung between the brood frames so that the strips contacted as much of the brood as possible. The strips had an uncoated section at the top, which was bent at a 90° angle to the rest of the strip and allowed the strips to hang without falling through the frames.

Thymol Florist Blocks. Whittington et al. (2000) showed efficacy against varroa using vermiculite blocks dosed with 3.8 g of thymol, for a two-hive body colony. Because our experiment used colonies one hive body in size, the dose was halved. To achieve a release rate of 0.15 g of thymol per hive body per day from the florist blocks at 33°C, 1.9 g of thymol was applied to the blocks. Pieces of vermiculite block 40 x 55 x 7 mm were treated with 19 ml of a 10% thymol in hexane solution. The thymol/hexane solution was poured over the florist block and the hexane allowed to volatilize, leaving the thymol behind. Florist blocks were weighed before the application of the thymol/hexane and after the hexane had evaporated. Loss of thymol during the transfer from hexane to florist block, and the subsequent volatilization of the hexane, did not significantly reduce the amount of thymol being applied to the blocks. One thymol block was placed in each colony, centered on top of the frames under the inner cover.

Magic 3 Florist Blocks. Examining the LD₅₀ values for thymol and Magic 3 reported by Lindberg et al. (2000), 16.4 ml of Magic 3 is equivalent to 3.8 g of thymol in efficacy, and a release rate of 1.2 ml of Magic 3 per day is equivalent to 0.31 g of thymol per day. As used in this experiment, Magic 3 had a release rate from vermiculite blocks of 0.6 ml per day for 4 days using an initial dose of 16.4 ml of Magic 3 on the block; after this time the release rate was variable (unpublished data). From these data we calculated that a release rate of 0.6 ml per day was optimal for a one-hive body-sized colony. This rate was achieved by pouring 16.4 ml of Magic 3 onto the florist blocks and allowing it to absorb. All florist blocks were enclosed in wire mesh in the colonies to prevent the bees from chewing them, and were centered on top of the frames under the inner cover.

Wick System. 50 ml of a 1.2% thymol in canola mixture was poured into an LDPE sleeve and absorbed by a 1.27 cm cotton wick, which was sewn into one end of the sleeve. After the wick had absorbed the 50 ml, an additional 350 ml of the 1.2% thymol in canola mixture was poured into the LDPE sleeve, which was used as a reservoir for the fluid. The opposite end of the sleeve was closed off with a paper clip and the reservoir was placed horizontally centered over the tops of the frames with the wick dropping down between the frames in the brood rearing area.

Experiment 2.

Colonies. To select colonies for the experiment, 45 two hive body-sized colonies from existing Simon Fraser University apiaries were sampled for varroa using a 24 h natural mite drop test. The 30 colonies with the highest mite infestations were used for the experiment and were randomly assigned to treatments. Colonies were not equalized for brood or adult bees. These colonies also were seeded with 200g of bees from colonies

that were known to be highly infested from a separate apiary to ensure that mites were present. All experimental colonies were then moved to an apiary at the British Columbia Ministry of Agriculture in Abbotsford B.C. The experimental treatment period began on 2 September 1999 and ended 25 September 1999. Colonies were treated with the antibiotic oxytetracycline hydrochloride (Oxytet-25, Medivet Pharmaceuticals, High River, Alberta, Canada) during the course of the experiment to prevent American foul brood disease, and were fed approximately 5.7 liters of sugar syrup in a two parts sugar to one part water concentration. The sugar syrup contained the antibiotics oxytetracycline hydrochloride (Oxytet-25, Medivet Pharmaceuticals, High River, Alberta, Canada) and fumagilin (Fumagilin B, Medivet Pharmaceuticals, High River, Alberta, Canada). Mites were sampled from the colonies in the manner described previously, except sheets were pulled and replaced every eight d for 24 d. Treatments also were replaced every eight d.

Treatments. Ten colonies were used in each of two treatments and a control. Three LPDE sleeves (described in experiment 1) per hive body were placed in colonies with 16.4 ml of Magic 3 per sleeve. For the first treatment, ten colonies had the sleeves hung with loops of wire between the outer frames. Sleeves were alternated so that in the top hive body two sleeves were located between frames one and two and one sleeve was between frames nine and ten, and in the bottom hive body the sleeve arrangement was reversed (Fig. 1B). In the second treatment, ten colonies had three sleeves per hive body with the same dose of Magic 3, 16.4 ml. These sleeves were placed horizontally across the top bars of the frames directly over the brood rearing area in both the top and bottom hive body (Fig. 1C). The remaining ten colonies were used as an untreated control and were left undisturbed.

Results

Experiment 1.

Varroa Mite Efficacy. There were significant differences in efficacy of the treatments in killing varroa ($F = 11.9871$; $df = 5, 42$; $P < 0.001$). The Magic 3 LDPE sleeve was significantly more effective in the control of varroa mites than any of the other treatments (Fig. 2). None of the treatments except for the Magic 3 LDPE sleeve were more effective than the untreated control ($P < 0.05$).

Tracheal Mite Efficacy. Tracheal mite levels were similar among colonies at the beginning of the experiment ($F = 0.3003$; $df = 5, 42$; $P = ns$). At the end of the experiment, significant differences were found for the Magic 3 sleeve (92% decrease from pre-treatment to post-treatment infestation levels), the Magic 3 florist block (80% decrease), and the thymol-canola wick (75% decrease) all having significantly lower tracheal mite levels than the untreated control (43% increase) ($F = 3.4525$; $df = 5, 39$; $P = 0.0112$). Neither the calcium carbonate/Magic 3 paste (43% decrease), nor the thymol florist blocks (58% decrease) were significantly different from the other treatments (Fig. 3).

Post-treatment colony size. At the beginning of the experiment there were no differences in colony population with respect to brood area ($F = 0.1654$; $df = 5, 42$; $P = ns$) or adult bee population ($F = 0.5165$; $df = 5, 42$; $P = ns$). All brood areas increased except for the Magic 3 LDPE sleeve which had little change from the beginning to the end of the experiment, and significantly less brood than the Magic 3 florist block and the

untreated control at the end of the experiment ($F = 2.7792$; $df = 5, 42$; $P = 0.0295$) (Fig. 4A). For all treatments the adult population increased over the course of the experiment, but the Magic 3 LDPE sleeve had fewer adults (although not significantly less) than the other treatments (Fig. 4B).

Experiment 2.

Varroa Efficacy. There were no significant differences in the efficacy of the treatments in killing varroa ($F = 0.7446$; $df = 2, 25$; $P = ns$). None of the treatments were significantly different from the untreated control (Fig. 5A).

Post-treatment colony size. There was no difference between the colonies in colony size at the beginning of the experiment with respect to brood area ($F = 1.2459$; $df = 2, 27$; $P = ns$). The brood area decreased in all treatments, but no treatment was significantly different than the others ($F = 1.2964$; $df = 2, 27$; $P = ns$) (Fig. 5B).

Figure 1. Treatment placement for Magic 3 sleeves in experiments one (A) and two (B and C). "Top hive-body" and "Bottom hive-body" represent the two hive bodies that make up a colony and their relative positions in the colony. "Frame number" refers to the position of the frame in the colony and can be used as a reference point for the placement of the sleeves.

Fig. 1

Treatment Placement

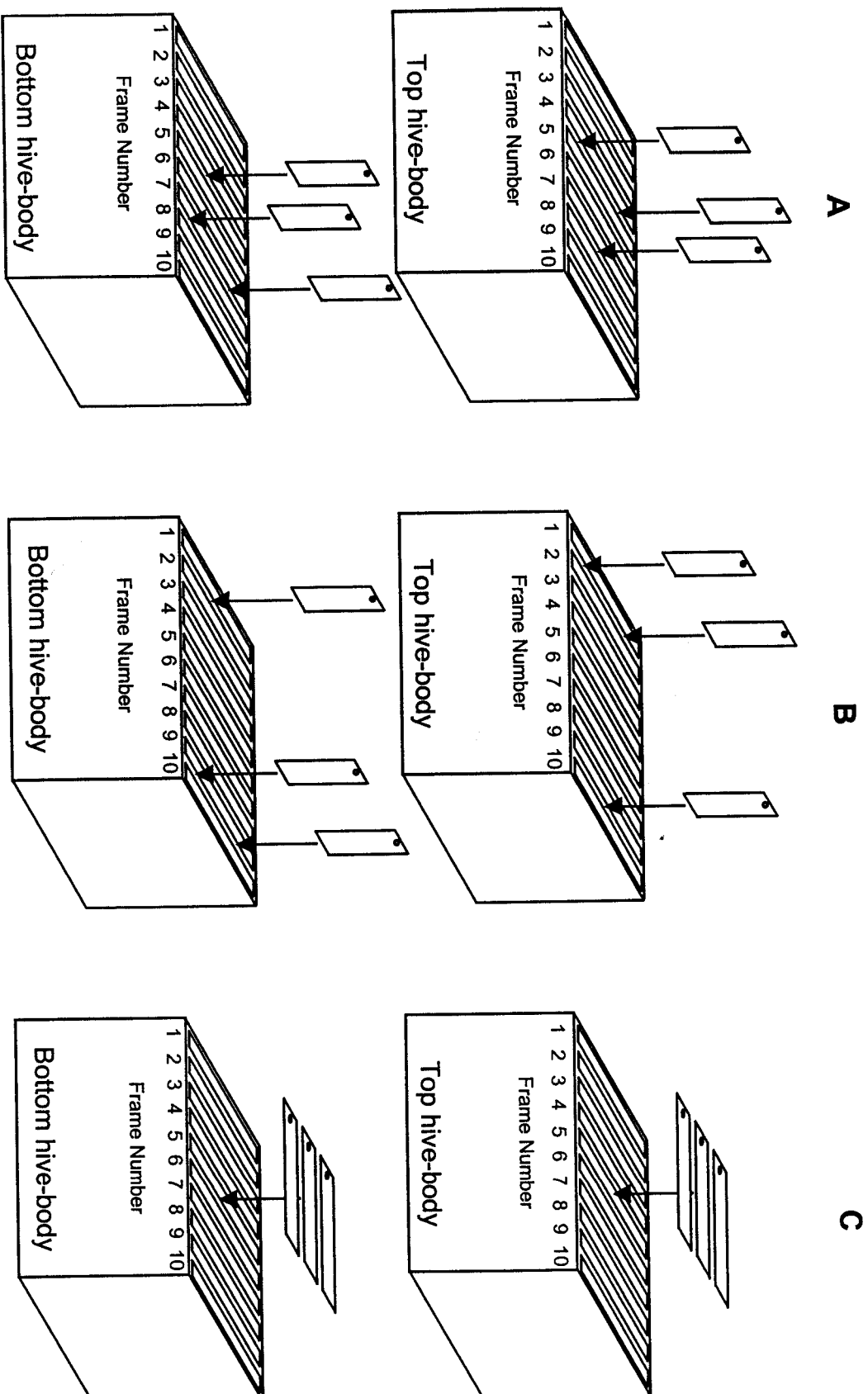


Figure 2. The relative control efficacy of each treatment against *V. destructor*. Means with the same letters are not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 2

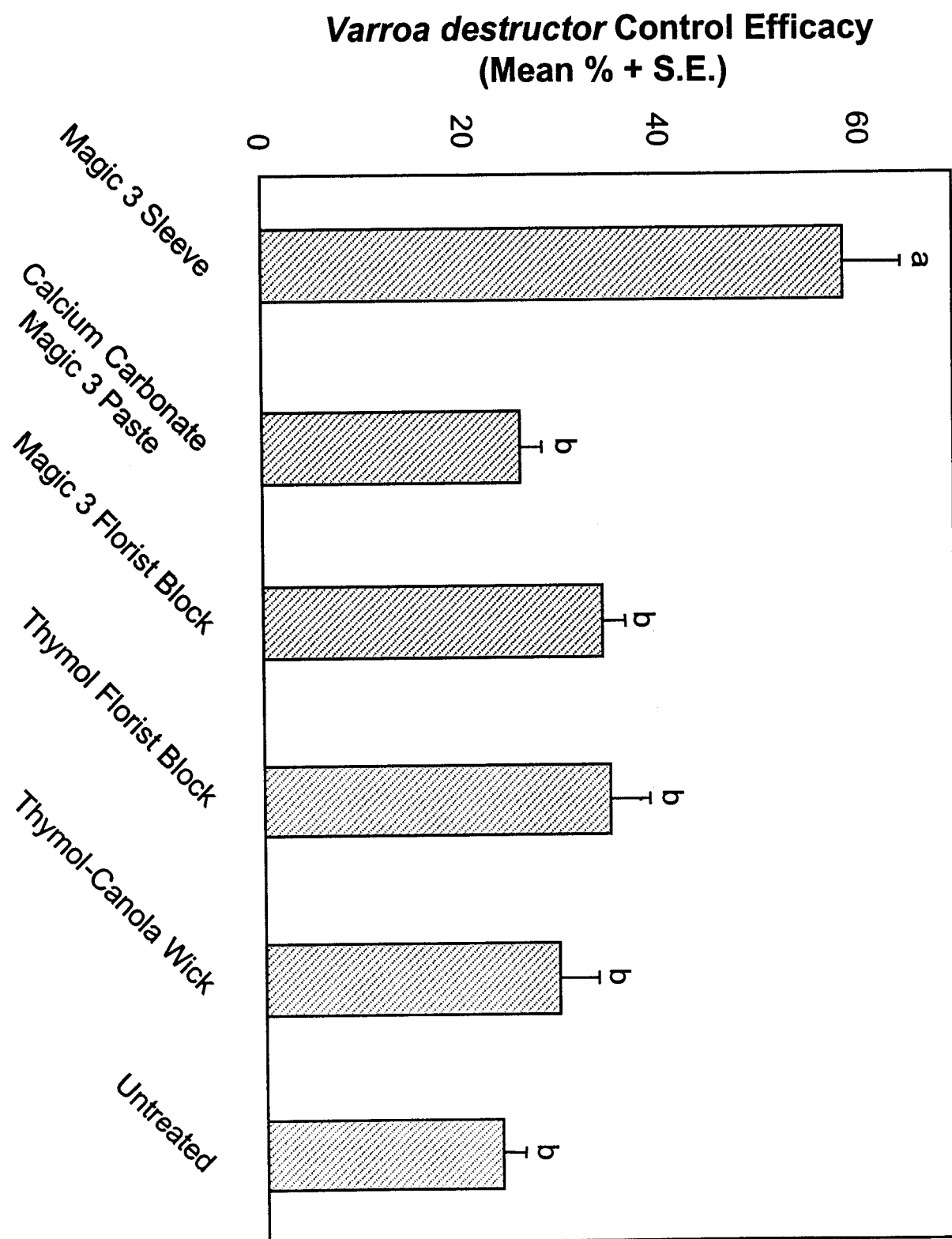


Figure 3. Pre and Post-treatment *A. woodi* (Tracheal mite) infestation levels. Means with the same letters are not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 3

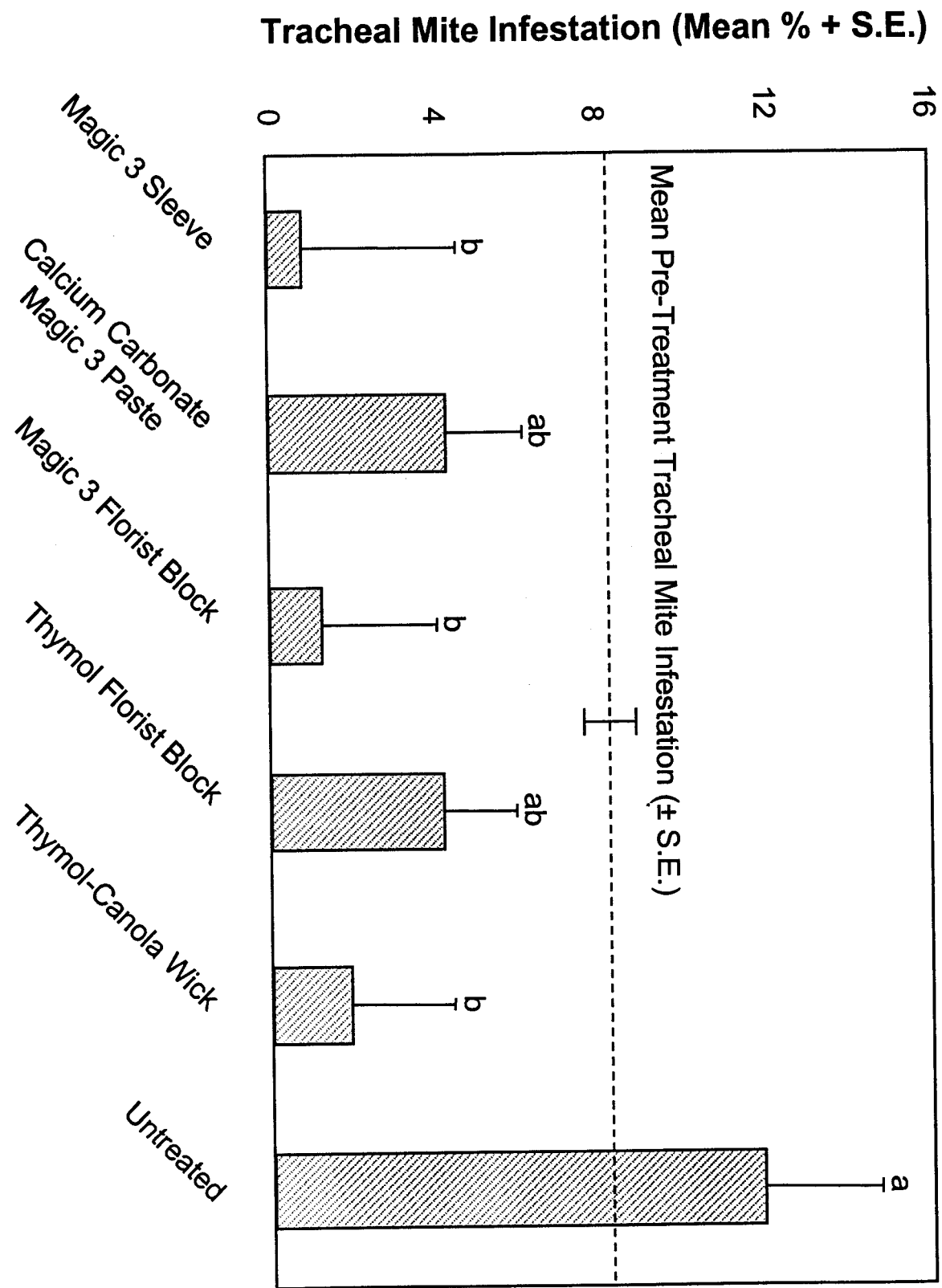


Figure 4. Sealed brood area (A) and adult population (B) of treatment groups at the end of the treatment period, compared with pre-treatment brood area and adult population. Means with the same letters are not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 4

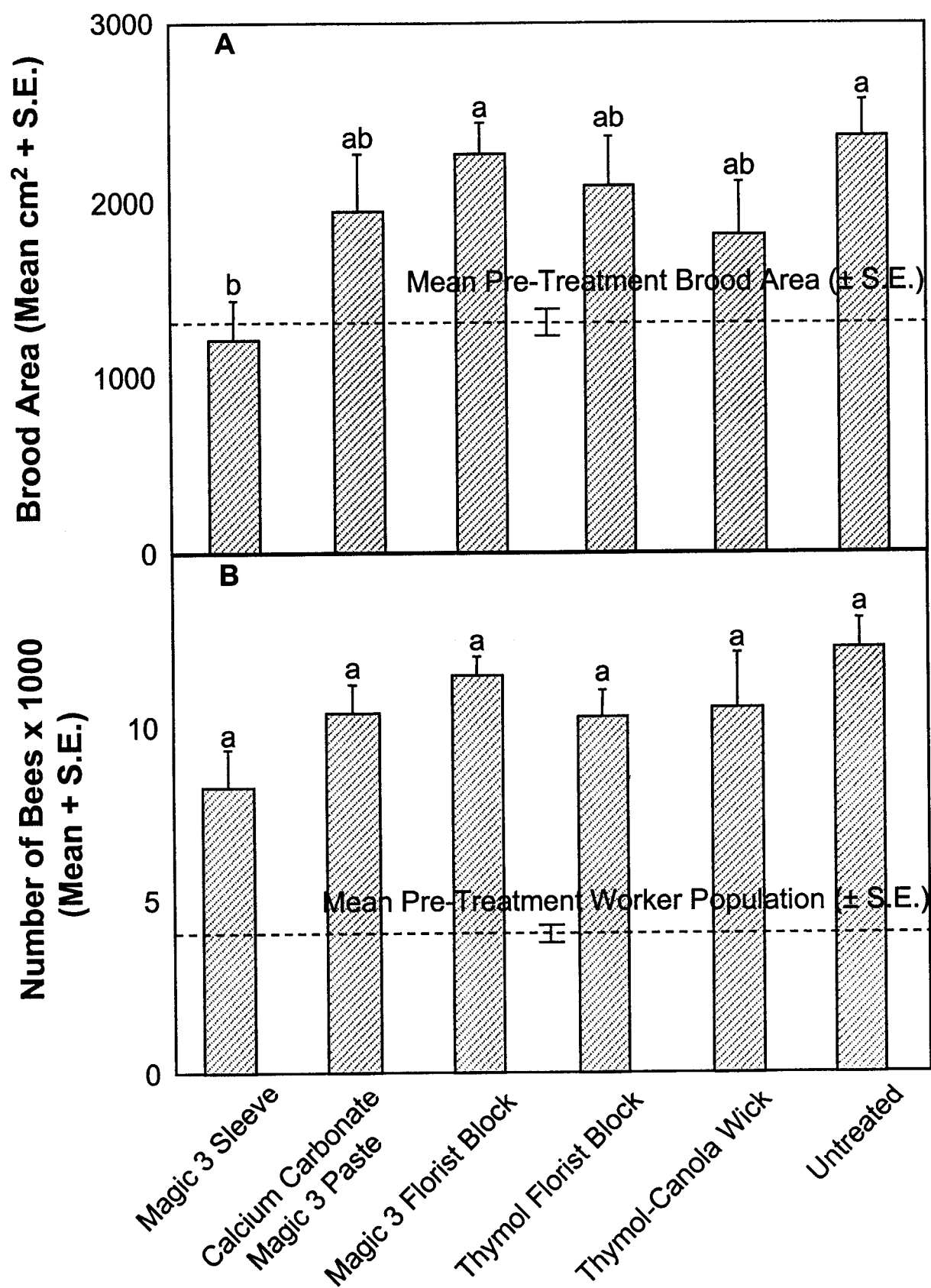
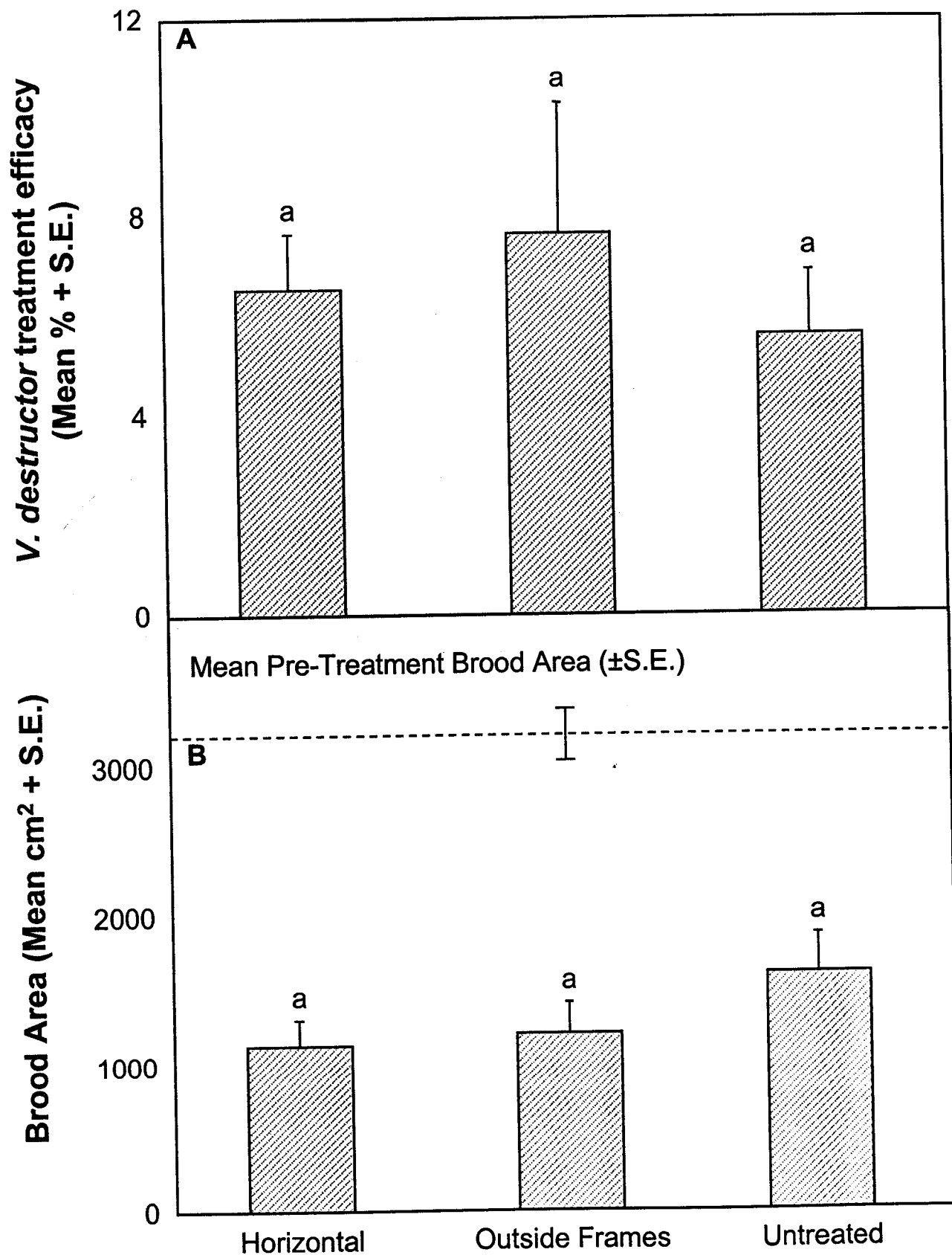


Figure 5. The relative control efficacy of each treatment against *V. destructor* (A) and the brood area (B) at the end of the treatment period in comparison to pre-treatment brood area. Means with the same letters are not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 5



Discussion

Magic 3 showed the highest varroa control when administered as a sleeve treatment during the first experiment, but these higher levels of control were offset by decreased adult populations and significantly decreased brood production. Subsequent tests in experiment two with sleeve placement showed that altering the location of the treatment decreased brood mortality, but lowered the capacity of Magic 3 to kill mites so severely as to render the product ineffective in those locations. It is not clear whether Magic 3 itself or simply physical interference from the sleeve placed next to the brood caused the drop in brood production.

The variability in treatment efficacy found in previous studies with botanical compound applications also was present in our results. The standard (positive control) of the thymol block (Calderone et al. 1997, Whittington et al. 2000) showed 35% control of varroa mites, which is at the lower end of the 30% to 70% range found in previous studies (Calderone et al. 1997, Whittington et al. 2000).

Tracheal mite control with Magic 3 and with thymol infused canola oil was dramatic, and either of these applications could be used given the proper dosage and placement. Future experiments should examine a range of Magic 3 doses either in the field or in a bioassay system similar to the one used by Lindberg et al. (2000) in order to determine the dosage required to provide tracheal mite control. A thymol/canola oil mixture administered as a spray can control tracheal mites but kills bees (Whittington et al. 2000). The same thymol/canola oil mixture was put into the wicks in our experiments and also showed good efficacy against tracheal mites without the detrimental effects to

bees and brood of a spray. In both experiments bees were able to come into direct contact with the mixture, and future study of bees' interactions with a thymol/canola oil release device would be useful.

Magic 3 has better potential to control both tracheal and varroa mites than thymol alone, but only if placement and release technology can be developed to reduce negative effects on brood. The optimal release device should administer a constant dose over a set period of time and be easy to apply. The LDPE sleeves tested here were a first step towards refining this release technology. These can be applied similarly to Apistan strips and provide a specified dose for up to 8 days. A single reservoir releasing a higher dose per day might be more useful if the placement can be optimized.

Chapter III

Integrated Pest Management for Varroa Control

Integrated Pest Management (IPM) has the potential to reduce chemical use and prolong the life of existing pesticides. IPM utilizes multiple control methods to reduce the development of pest resistance to any single control, thus reducing the exposure time of the pest to the chemical, exploiting different biological pathways that provide control, and taking advantage of cultural control methods that reduce pest populations. The goal of IPM is not to eliminate chemical use but to reduce the amount of pesticides that need to be applied and increase the longevity of current control methods.

The varroa mite is well suited for IPM because a number of control methods are available, including chemical controls and cultural or genetic methods including modified bottom boards (Pettis and Shimanuki 1999), hygienic bee stock (Spivak 1996, Spivak and Reuter 2001), drone-brood trapping (Calis et al. 1999) and heat treatments (Huang 2000).

Modified bottom boards can reduce mite levels because varroa lives part of its life on the outside of adult bees as an external parasite. Some of these phoretic mites lose their grip and fall to the bottom of the colonies. 40-50% of the mites that fall onto the bottom board are still alive and search for a new bee host to carry them back up into the colony (Pettis and Shimanuki 1999, Jeff Pettis, United States Department of Agriculture, Beltsville, MD pers. comm.). This behavior can be exploited with a screened bottom board that prevents mites from interacting with bees as they return to the hive. Mites will not climb more than two inches before they cease climbing (Pettis and Shimanuki 1999). Therefore, if a screen is placed over a dead air space greater than two inches, mites that have fallen through the screen will be unable to re-enter the colony.

Hygienic behavior in bees was first described in the 1930s by O.W. Park, F.B. Paddock, and F.C. Pellet as a possible resistance mechanism to the bacterial infection American foulbrood (AFB) (Spivak and Gilliam 1998a). Since the exact cause of the resistance was not known, it was felt that nest-cleaning played a key part in AFB resistance. Research on AFB-resistant colonies was continued by A.W. Woodrow to determine the basis of resistance (Woodrow 1941), and further investigated by W.C. Rothenbuhler in the 1960s and 1970s.

In a series of papers, Rothenbuhler presented data from experiments where he tested four lines of bees, two resistant and two susceptible, to determine the heritability of hygienic behavior. He first tested the four lines to show that they actually were resistant or susceptible respectively, then performed a series of backcrosses of the F_1 generation with the original parent queens to determine that this behavior involved two loci.

Rothenbuhler assumed that the resistant trait was homozygous recessive, so following a Mendelian model the number of loci involved in the trait could be determined by looking at the percentage of offspring from the F_1 /parent cross; 50% and the trait involves only one loci, 25% and the trait involves two - a truly elegant experiment to determine the genetics of this behavior. He found that 21% of the backcrossed offspring displayed the trait, so he assumed that the trait involved two loci but was unsure whether this model explained the reason for resistance entirely (Rothenbuhler 1964a, Jones and Rothenbuhler 1964, Thompson 1964, Rothenbuhler 1964b). Later experiments showed that environmental factors such as previous experience (i.e. age) of the bee, colony demography, and resource availability also affect hygienic behavior expression and may

have been the cause of some of the variability seen in the earlier genetic experiments (Trump et al 1967, Momot and Rothenbuhler 1971).

Research on AFB resistance in bees continued through the 1960s and 1970s but ceased when antibiotics began to be used heavily for the control of AFB. Antibiotics were easy to apply and provided consistent, satisfactory results. Interest in this behavior has been renewed recently due to the discovery of antibiotic-resistant strains of the bacterial disease (Miyagi et al. 2000). Hygienic behavior also has the potential for control of the fungal disease chalkbrood, the bacterial disease European foulbrood (EFB), and varroa mite suppression (Oldroyd 1996, Spivak and Gilliam 1998b).

Hygienic behavior is heritable so it can easily be selected for and bred into bee stocks (Kulincevic et al. 1992, Harbo and Harris 1999). Screening for hygienic behavior involves determining the response time of worker bees to remove dead brood. All bees will remove dead brood over a period of time, but bees expressing hygienic behavior remove brood more quickly, and dead brood removal is positively correlated to disease reduction and removal of varroa infested brood (Spivak 1996, Boecking and Spivak 1999, Palacio et al. 2000). The positive traits bred for by beekeepers, such as low aggression and high honey productivity, are not compromised when selecting for hygienic behavior. Hygienic colonies, when compared to unselected commercial colonies, had significantly less chalk-brood, no American foulbrood, produced significantly more honey, and had significantly fewer varroa mites (Spivak 1996, Spivak and Reuter 1998, 2001).

The use of modified bottom boards can remove 15-30% of the mite population (Pettis and Shimanuki 1999), thymol can kill anywhere from 30 to 90% of the mite population with an average of about 70% control, and hygienic bees can lower a mite

population in a lightly infested colony by 40-60% (Spivak 1996, Spivak and Reuter 2001). Each of these control methods used alone may not be sufficient, but the control provided by a combination might maintain mite populations below an economically damaging threshold. While many researchers have commented on and encouraged the idea of IPM for varroa control (i.e. Tangkanasing et al. 1988, Manino et al. 1996, Sammataro and Needham 1996, Ellis Jr. 2001), a long-term experiment determining how varroa would be affected by an IPM strategy has not yet been attempted.

Our experiment compared the use of thymol, modified bottom boards, and hygienic bee stock together to variously timed applications of Apistan, thymol, or Apistan and thymol. Modified bottom boards and hygienic bee stock were chosen because they can be incorporated into colony management easily and require only a small amount of extra maintenance. Thymol was chosen because of its performance in previous experiments and its regulatory acceptability as a food-grade botanical compound.

Many pest management regimes rotate different chemicals to relieve selection pressure on the pest to any one pesticide. I applied this principle to varroa control to determine if thymol could be used in conjunction with the synthetic chemical control Apistan. Thymol does not reduce mite populations to the same extent as Apistan, but perhaps the level of control provided by thymol rotated with Apistan would keep mite populations below their economic threshold. Mite controls typically are applied to colonies in the early spring before colonies begin producing honey and in the late summer/early fall before colony size reduction begins in preparation for overwintering, so we also tested the timing of Apistan/thymol treatments.

An IPM approach does not have to be non-chemical, and the main component of any IPM system is the use of a combination of control methods. Given this definition, the rotation of Apistan and thymol used in this experiment qualifies as an IPM approach that can be classified as "rotational IPM" or RIPM. The IPM approach involving hygienic bees (a genetic control), modified bottom boards (a cultural control), and thymol application (a chemical control), can be classified as a "mixed IPM" or MIPM approach where methods from several disciplines are combined in one holistic system.

The objective of this experiment was to determine how colonies would perform when being treated for varroa with either a mixed IPM treatment, a single chemical treatment such as Apistan alone or thymol alone, or Apistan and thymol combined in a rotational IPM treatment. Colony performance was evaluated by examining colony survival, adult bee population, brood population, honey production, and mite infestation levels.

Materials and Methods

Assessments. The adult and brood populations of all test colonies were assessed the week before the treatments (Burgett and Burikam 1985). Sticky boards were placed on the bottom boards of all colonies before and after treatment to capture mites for a 48 h period to determine colony mite loads, except for the fall of 1999 when mites were assessed only before the treatments were placed on the colonies. Colony survival was monitored throughout the course of the experiment. Honey production was measured by

taking the weight of each hive body used for honey collection before and after the honey was extracted. The weight difference was considered as the amount of harvestable honey.

Data Analysis. Unless otherwise stated, means of all mite, adult bee, brood, and honey data were compared using the Tukey-Kramer HSD multiple comparison test ($\alpha = 0.05$) (JMP IN, SAS Institute Inc. 1997). Colony survival data was analyzed using a Product-Limit Survival Fit analysis ($\alpha = 0.05$) (JMP IN, SAS Institute Inc. 1997).

Treatments. The five treatments examined and their timing were:

<u>Spring Treatment</u>	<u>Fall Treatment</u>	<u>Sample Size (Colonies)</u>
Apistan	Apistan (Single Chemical)	8
Thymol	Thymol (Single Chemical)	9
Apistan	Thymol (RIPM)	9
Thymol	Apistan (RIPM)	9
MIPM	MIPM	10

Colonies. Treatments were assigned randomly to all colonies except for the MIPM colonies, which had to be established with hygienic queens and could not be randomized. The thymol was replaced at four-d intervals for 24 d. The Apistan was left on for the package specified six-wk period. The MIPM colonies were treated with three control methods simultaneously: 1) thymol; 2) modified bottom boards; and 3) hygienic queens. Modified bottom boards and hygienic queens were not removed from the colonies for the duration of the experiment.

Thymol. All colonies with a thymol component to the treatment were administered thymol blocks prepared in the manner described in Chapter II. However since colonies in this experiment consisted of two standard Langstroth hive bodies instead of one, the dose applied was doubled from 1.9 grams to 3.8 grams.

Screened Bottom Boards. A screened bottom board with a two inch space between the screen and the bottom of the board was used to prevent mites from crawling back into the colony. Since this board is not accessible to bees, it also prevents the mites from reattaching themselves to the bee and gaining access back into the colony.

Hygienic Behavior. Colonies from Simon Fraser University apiaries were screened for hygienic behavior using a method in which freeze-killed brood were placed in a colony (Spivak 1996). The length of time required for adult bees to remove the dead brood was assessed. If 90-95% of the brood was removed in a 48h period on two consecutive tests, the colony was considered hygienic.

To freeze-kill the brood, a metal cylinder with a diameter of approximately 8 cm was pressed into an area of sealed brood. The frame containing the brood was placed on the ground horizontally. Liquid nitrogen was poured into the cylinder and left until all of the brood within the area had been frozen solid. The cylinder then was removed from the frame leaving the frozen brood behind, the area of frozen brood was marked with an arrow on the top of the frame, and the frame was returned to the colony. Twenty-four hours later the frame was removed again and the percentage of brood cleared from the 8 cm patch was recorded. If more than 75% of the brood had been removed, the test was repeated in the same colony on a different frame of brood. If the bees in the colony removed more than 75% of the dead brood in two

consecutive 24 hour periods, the colony was considered hygienic. This modified screening method was used to expedite the process.

In order to produce queens whose offspring would display hygienic behavior, virgin queens from colonies displaying the highest level of hygienic behavior were mated with drones from other hygienic colonies and subsequent generations were tested for hygienic behavior. These matings were conducted at an isolated mating site. New queens were screened for hygienic behavior and available for field-testing by fall 1999.

Results

Fall 1999 colony assessments at the beginning of the experiment are summarized in Figure 6. There were no significant differences between treatments in any of the colony conditions measured.

Colony Survival. There was no significant difference in colony survival over the course of the experiment (Log-Rank Chi Square = 6.1246; df = 4; P = ns; Wilcoxon Chi Square = 6.0010; df = 4; P = ns) (Fig. 7).

Post-treatment colony population. There were no significant differences between treatments in adult bee levels in the fall of 1999 ($F = 0.7603$; df = 4, 40; P = ns), in the spring of 2000 ($F = 0.5091$; df = 4, 35; P = ns), in the fall of 2000 ($F = 2.1873$; df = 4, 32; P = ns), and in the spring of 2001 ($F = 0.0901$; df = 4, 17; P = ns) (Fig. 8). There also were no significant differences between treatments in brood

levels in the fall of 1999 ($F = 0.9790$; $df = 4, 40$; $P = ns$), in the spring of 2000 ($F = 0.3435$; $df = 4, 34$; $P = ns$), in the fall of 2000 ($F = 0.3929$; $df = 4, 32$; $P = ns$), and in the spring of 2001 ($F = 0.7508$; $df = 4, 17$; $P = ns$) (Fig. 9).

Varroa Mite Efficacy. There were significant differences in mite levels only after treatment in the fall of 2000 ($F = 10.0042$; $df = 4, 30$; $P = <0.0001$) (Fig. 10). The Apistan/Apistan and thymol/Apistan treatments had significantly fewer mites than the Apistan/thymol and thymol/thymol treatments, but not significantly fewer mites than the mixed IPM treatment. There were no significant differences in mite levels at any other time; fall of 1999 ($F = 1.4021$; $df = 4, 40$; $P = ns$), before treatment in the spring of 2000 ($F = 0.8141$; $df = 4, 33$; $P = ns$), after treatment in the spring of 2000 ($F = 2.5489$; $df = 4, 35$; $P = ns$), before treatment in the fall of 2000 ($F = 1.0682$; $df = 4, 31$; $P = ns$), before treatment in the spring of 2001 ($F = 2.6481$; $df = 4, 17$; $P = ns$) or after treatment in the spring of 2001 ($F = 2.26$; $df = 4, 15$; $P = ns$).

Honey Production. There were no significant differences between treatments in honey production ($F = 0.2105$; $df = 4, 33$; $P = ns$) (Fig. 11).

Figure 6. Adult bee population, brood population, and mite infestation level at the beginning of the experiment in the fall of 1999. No significant differences between treatments were found for any of the three variables (Tukey's HSD, $\alpha = 0.05$).

Fig. 6

Fall 1999 (Experiment Start) Variable Measurements

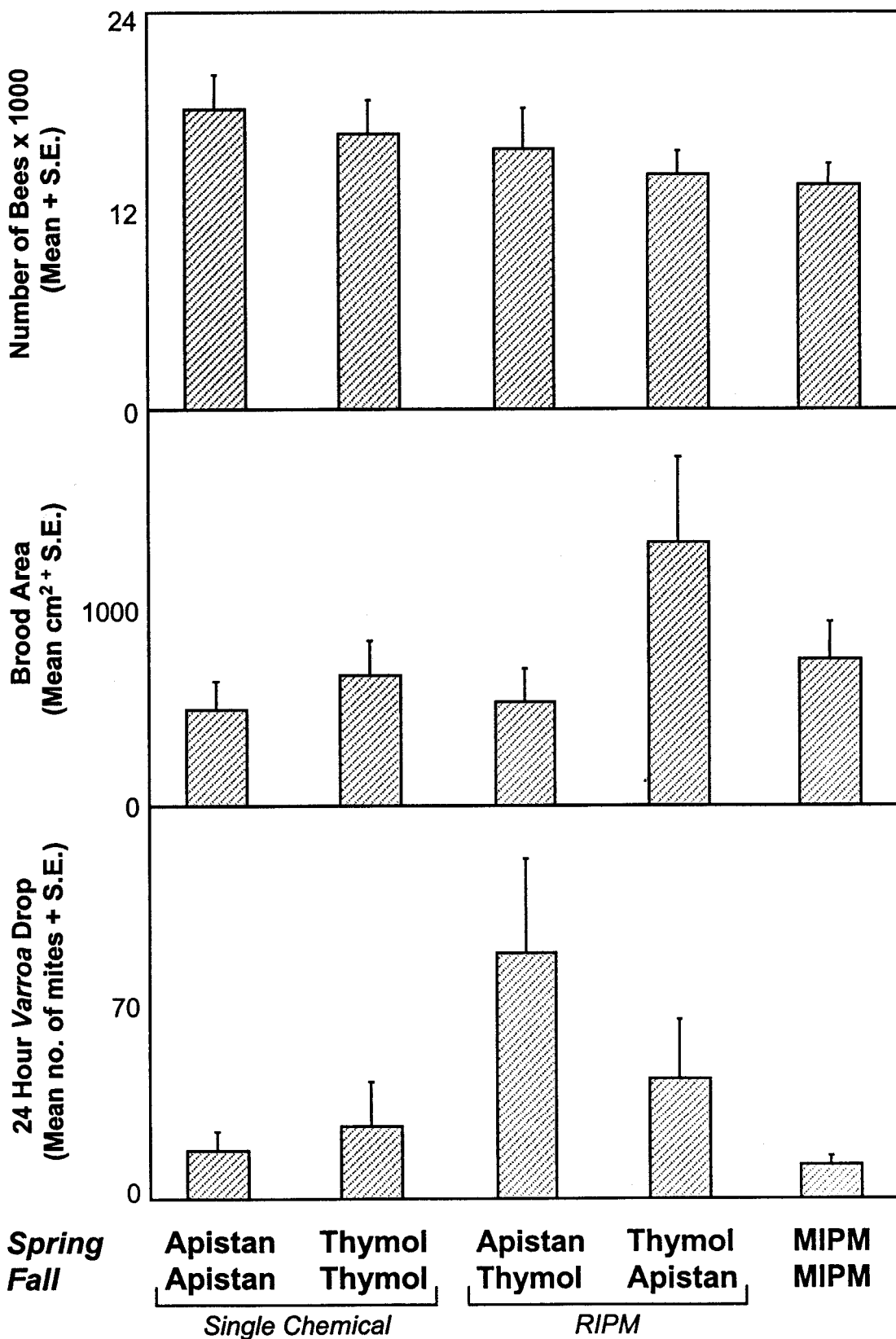


Figure 7. Colony survival over the treatment period. No significant differences were found between treatments (Log-Rank Chi Square and Wilcoxon Chi Square, $\alpha = 0.05$).

Fig. 7

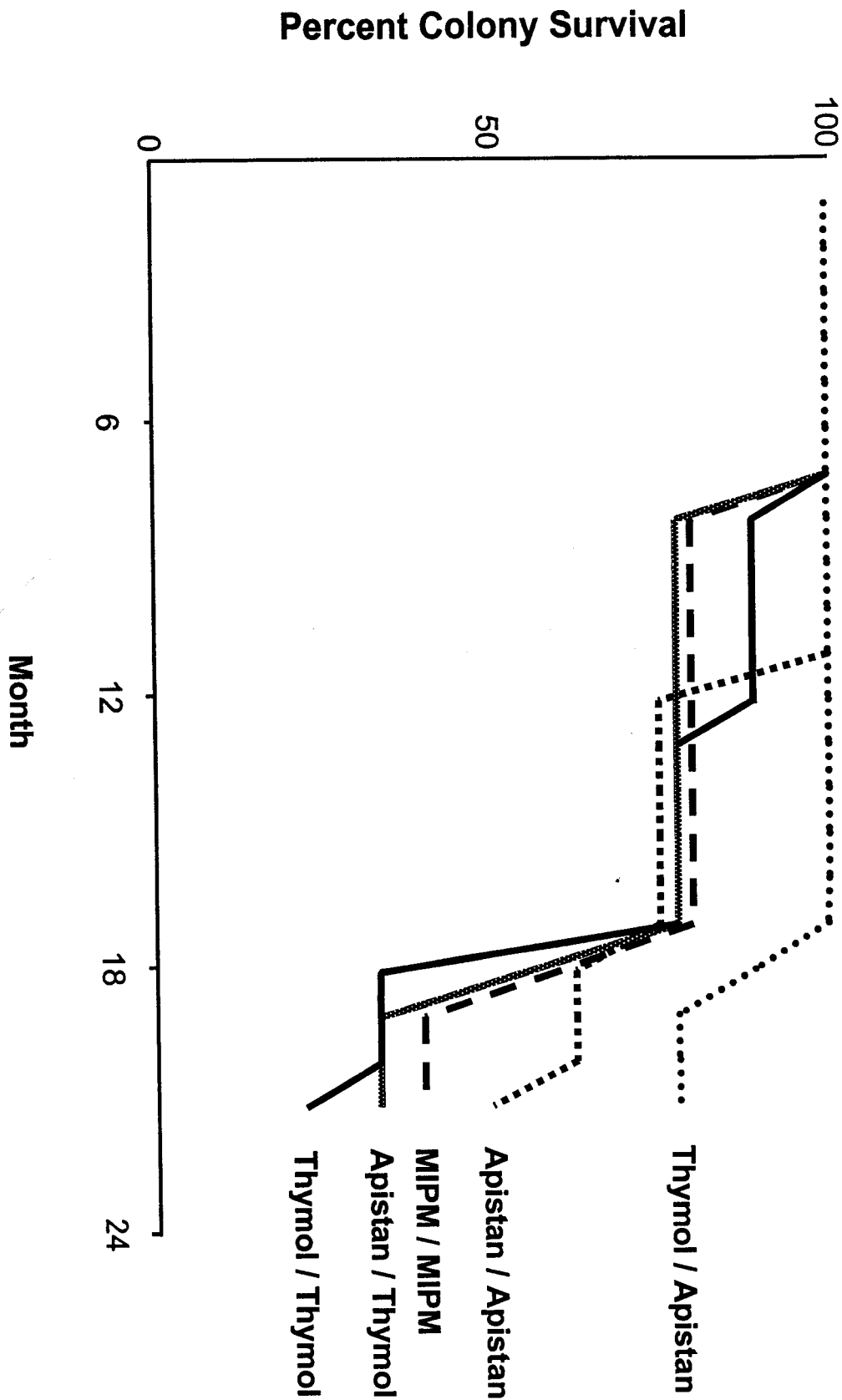


Figure 8. Adult bee populations over the duration of the experiment. On each assessment date, no significant differences between treatments were found (Tukey's HSD, $\alpha = 0.05$).

Fig. 8

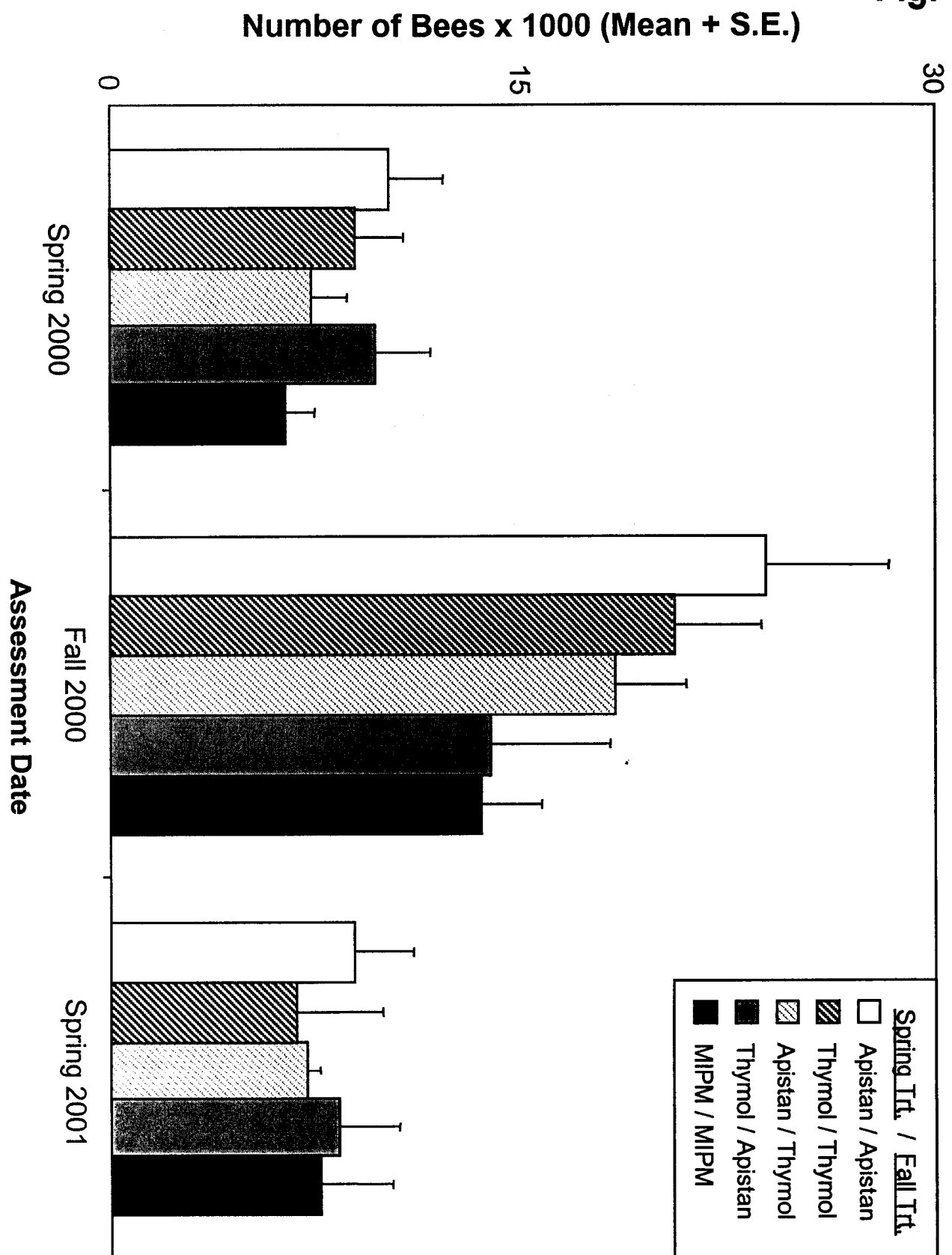


Figure 9. Brood populations over the duration of the experiment. There were no significant differences between treatments on each assessment date (Tukey's HSD, $\alpha = 0.05$).

Fig. 9

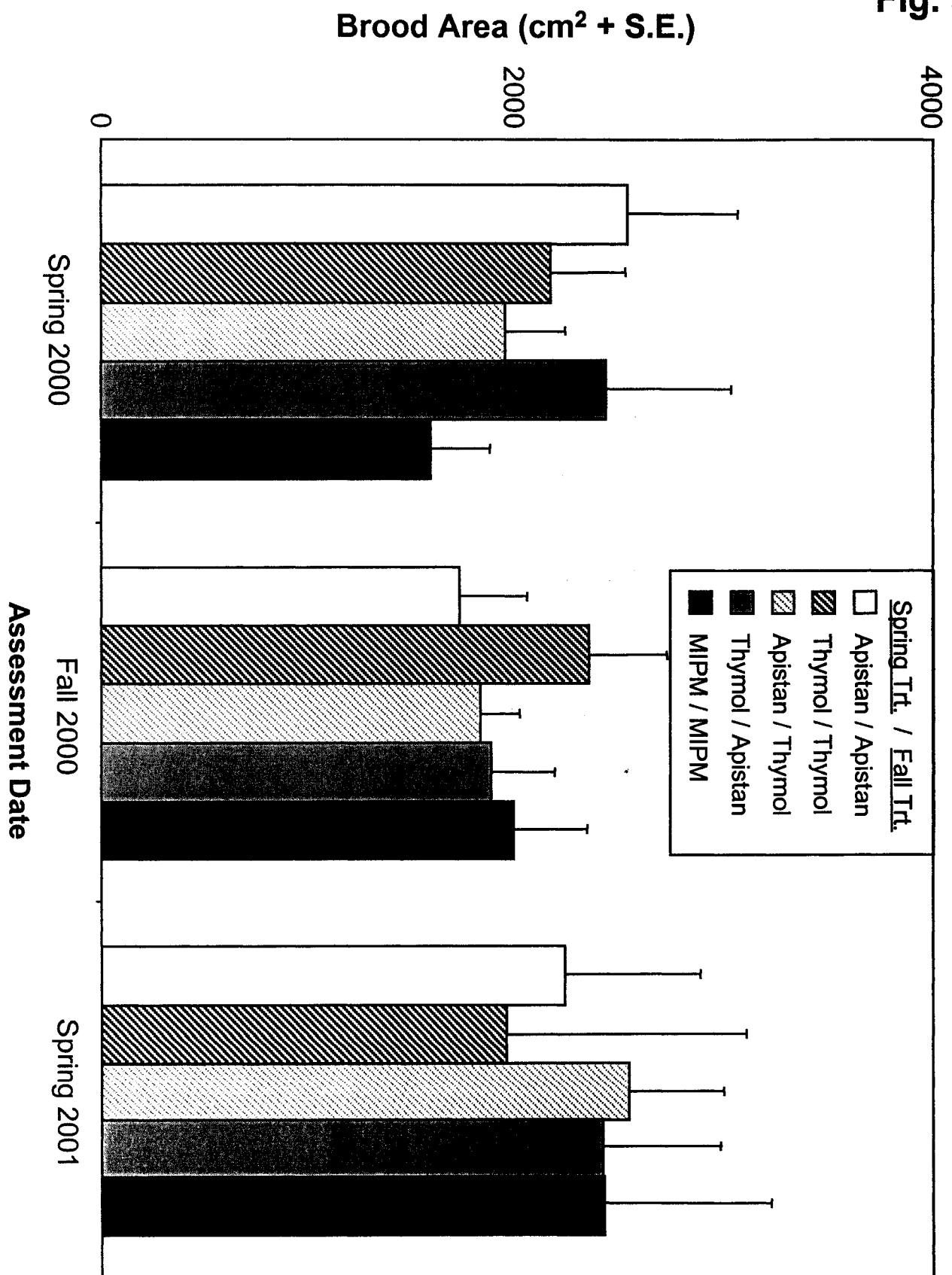


Figure 10. Varroa mite infestation levels (number of mites), as determined by a 24 hour natural drop, pre and post-treatment over the duration of the experiment. Where significant differences were found, means with the same letters indicate treatments that were not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 10

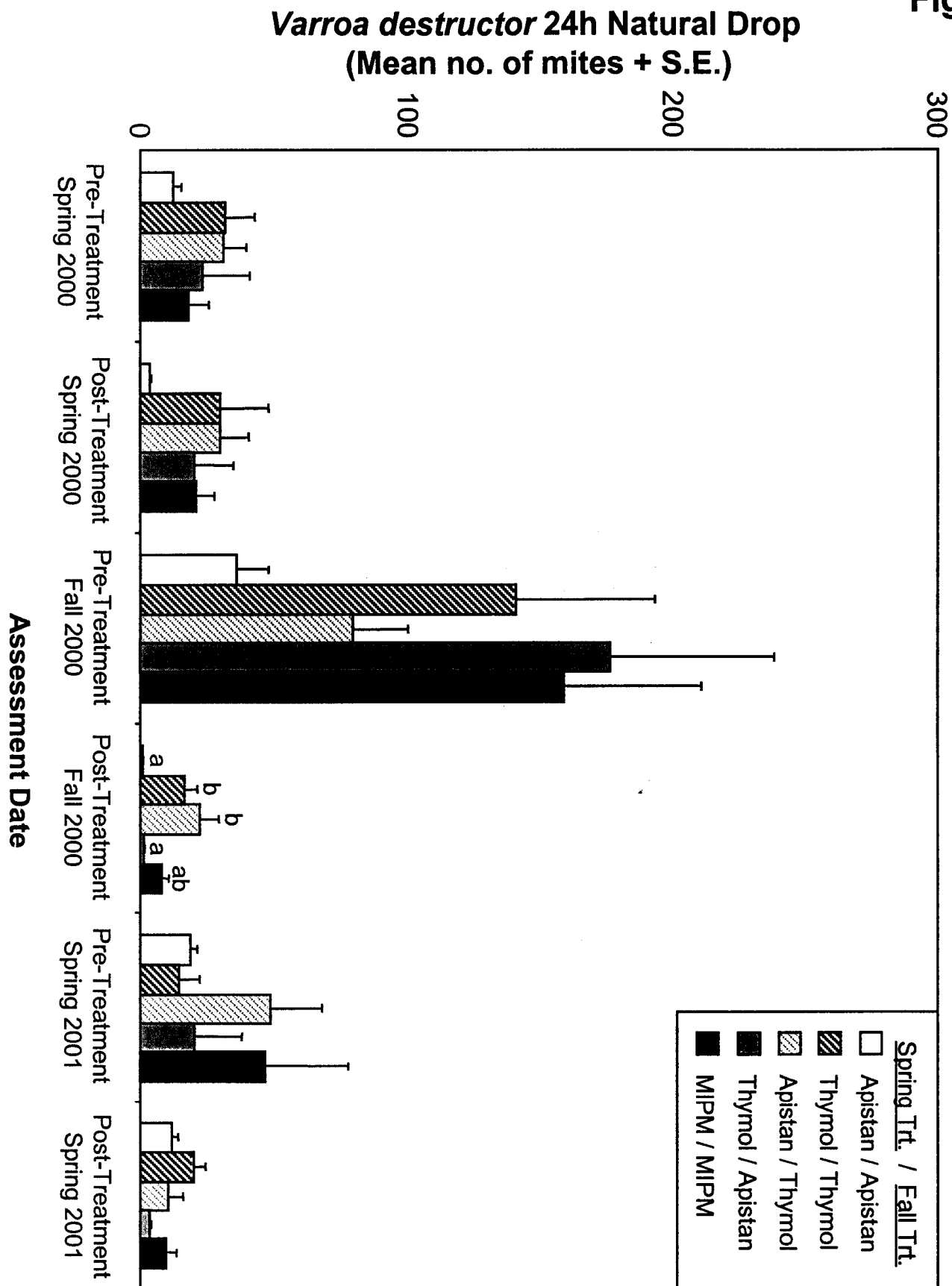
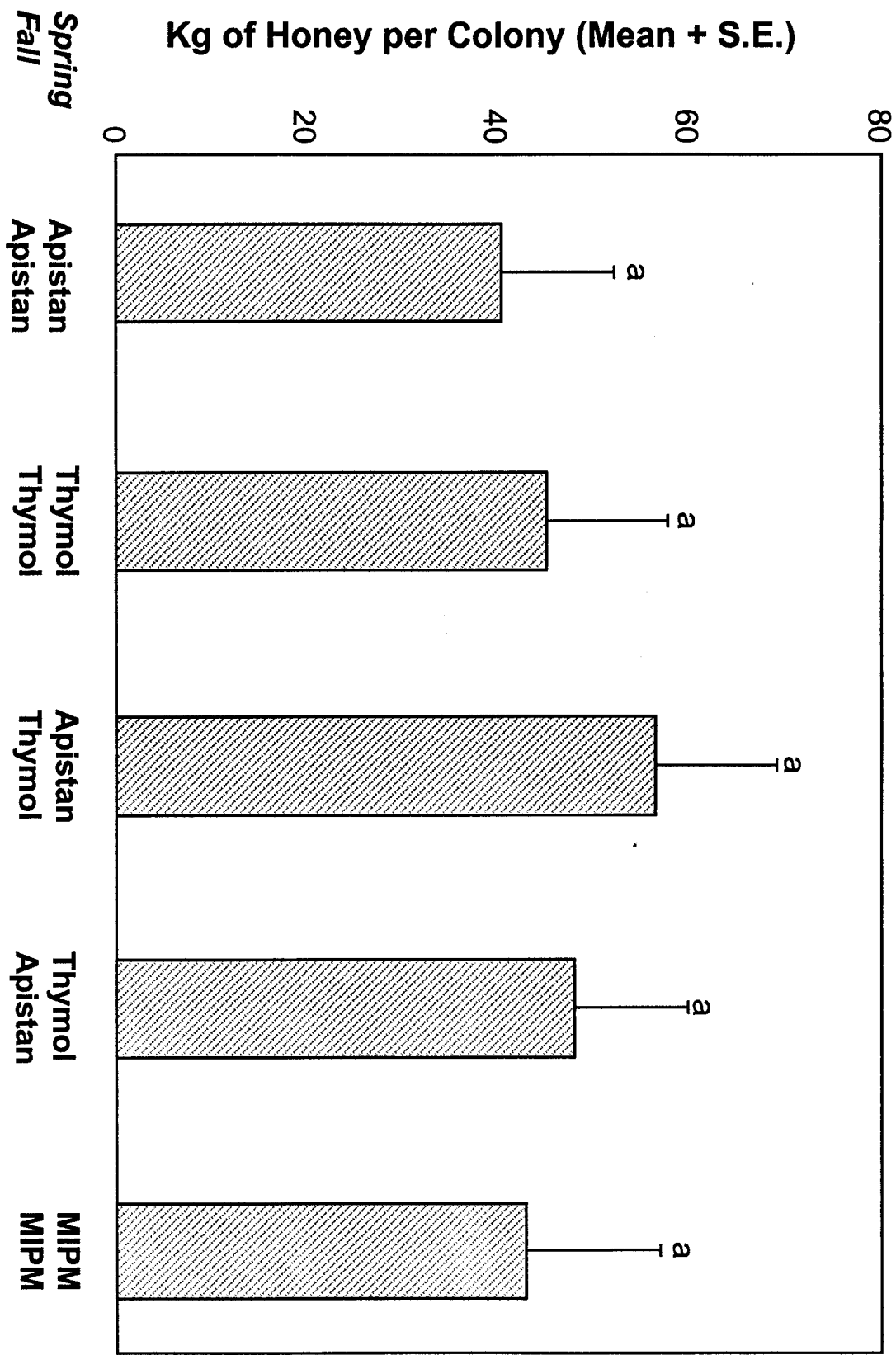


Figure 11. Honey production after the summer 2000 honey harvest. Means with the same letter were not significantly different (Tukey's HSD, $\alpha = 0.05$).

Fig. 11



Discussion

Comparisons of treatments throughout the experiment consistently showed no significant differences in adult bee and brood populations or in honey production, indicating that none of the treatments had adverse effects on the colony. Differences in mite populations did not appear until the latter half of the experiment, and then only in the fall. There was colony loss in all treatments, but there were no statistical differences in survival. However, there was enough colony loss in the thymol/thymol and Apistan/thymol treatments to suggest that thymol alone or used as a fall treatment when alternating with Apistan may not be recommended.

The use of Apistan or some other high (>90%) mite control product is important for fall treatment because of decreasing colony size. Overwintering bees not only have to stay alive for 3 - 4 months, but also rear the new generation of brood in the spring. If adult bees have been parasitized for the duration of the winter, they will be less capable of performing tasks in the spring. Further, as the winter bee population dwindles the mite population becomes concentrated on the remaining bees. If the brood infestation rate is high, the first generations of bees will be compromised, preventing the colony from growing or killing it outright.

The thymol/Apistan treatment exhibited the least colony loss and had significantly fewer mites than all other treatments except for Apistan/Apistan in the post-treatment mite drops tested in the fall of 2000. Thymol may provide adequate control in the spring when bee populations are growing, and can be used in rotation with Apistan, much as formic acid is used currently in Canada, to provide an alternate means of control and thereby reduce the likelihood of resistance developing.

Sixty percent of the MIPM colonies were lost over the course of the experiment, but most colony loss occurred at the end of the experiment. Therefore, applying Apistan

every other year might be useful to maintain an economically acceptable level of control while slowing resistance development.

Refining the administration of thymol in an MIPM or RIPM treatment such as thymol/Apistan could improve the mite control capabilities of both of these treatments. Variability in mite control has been a problem with thymol, but if a dependable, easy-to-apply treatment becomes widely available this variability can be reduced and thymol could take a more central role in varroa treatment.

The number of tools available for use in an IPM treatment is large enough to suggest that finding the best combination of treatments is an achievable goal. A non-chemical IPM treatment regime using modified bottom boards, heat-treatment, drone-brood trapping, and/or hygienic queens is possible, but the feasibility of such a treatment in terms of labor cost may be unreasonable and mite control efficacy may be too low. At the other extreme, U.S. beekeepers currently alternate Apistan in the fall and coumaphos in the spring, but the treatment cost, residue build-up, and chance for resistance development remains high (Elzen et al. 2001). A cost benefit analysis (Table 1) of each treatment suggests that combining cultural and chemical controls appears to be both feasible and desirable.

Thymol block cost was calculated from materials used to construct one thymol florist block and cage for one colony multiplied by three applications per treatment multiplied by two treatments per year (Sigma-Aldrich Canada, Oakville, ON.; Curtis Lumber, Burnaby, BC; Garden Works, Burnaby, BC). Formic acid cost was calculated from materials used to apply one dose of liquid formic acid multiplied by six applications per treatment multiplied by two treatments per year (JJ's Bee Supply, Cloverdale, BC). Modified bottom board cost was taken from the Mann Lake Ltd. Catalogue (Mann Lake Ltd., Hackensack, MN 2001). Coumaphos cost was calculated as the cost of the total number of coumaphos strips needed to treat one colony for one year (Mann Lake

Table 1. Varroa control efficacy, treatment, treatment component and labor cost (USD) on a per colony/per year basis for a selection of available control methods

	Annual treatment cost ^a	Annual labor cost ^b	Total cost	Varroa control efficacy (% mite drop)
Component				
Thymol Blocks	\$9.48	\$2.72	\$12.20	30-70%
Modified Bottom Boards ^c	\$0.75	\$0.07	\$0.82	15-30%
Hygienic Queens ^{cd}	\$2.00	\$0.00	\$2.00	40-60%
Formic Acid (Liquid)	\$2.02	\$3.24	\$5.26	70%-89%
Coumaphos	\$13.22	\$2.04	\$15.26	97%
Apistan	\$11.36	\$2.04	\$13.40	95-99% ^e
Treatment (This Study)				
Apistan/Apistan	\$11.36	\$2.04	\$13.40	
Thymol/Thymol	\$9.48	\$2.72	\$12.20	
Thymol/Apistan (RIPM)	\$10.42	\$2.38	\$12.80	
Apistan/Thymol (RIPM)	\$10.42	\$2.38	\$12.80	
MIPM	\$12.23	\$2.79	\$15.02	

A colony is a two hive body honey bee colony filled with bees and containing at least 10 frames of brood

^aTreatment cost is the cost of materials needed for one full year of treatment for one colony (i.e. 6 thymol applications. 2 Apistan applications, etc.; see text.)

^bLabor cost is the cost of one worker working for \$10 (USD) an hour applying treatments to one colony for one year.

^cModified bottom boards and hygienic queens do not need to be replaced every year. Therefore the cost was spread throughout the average life of each of the treatments, 10 years for a bottom board, and 2 years for a hygienic queen. Labor costs for replacing queens were considered as zero since most beekeepers generally requeen biannually regardless of queen type.

^dHygienic queens are available through various bee breeders but are usually open mated resulting in variable levels of hygienic behavior in the offspring. The cost shown here is the average difference between cost of a non-hygienic and hygienic queen taken from advertised rates in bee journals.

^eThe efficacy range given is for non-resistant bees. Apistan's efficacy range would be altered if resistance was taken into account.

Ltd., Hackensack, MN 2001). Apistan cost was calculated as the cost of the total number of Apistan strips needed to treat one colony for one year (JJ's Bee Supply, Cloverdale, BC).

Hygienic queen cost was more difficult to determine. Many bee suppliers offer queens that claim to have been screened for hygienic behavior, usually charging between \$9 and \$11(USD) for these queens, \$2 more on average than a queen not selected for hygienic behavior. However, since no certification process exists and the parentage of these queens is unknown, the true level of hygienic behavior displayed by the progeny of these queens may be variable. Labor costs for hygienic queens were considered as zero since the cost of applying the “treatment” is no different than the cost of requeening with a non-hygienic queen.

Labor costs were calculated as the cost of one laborer working alone. The time needed to apply and remove (where applicable) each treatment was determined, and the pay for a worker at ten dollars an hour to apply and remove one full treatment cycle to one colony for one year determined. Efficacy data were summarized from the literature (Calderone et al. 1997, Whittington et al. 2000, Kochansky and Shimanuki 1999, Hoppe et al. 1989, Pettis and Shimanuki 1999, Spivak 1996, Spivak and Reuter 2001, Baxter et al. 1999, Elzen et al. 1998).

The rotational IPM treatment, thymol/Apistan, provided the best control over the course of this experiment and also was the least expensive, followed closely by the single chemical Apistan/Apistan treatment and the mixed IPM treatment. Nevertheless, the MIPM regime may prove more desirable than the RIPM treatment in the long run for several reasons. Modified bottom boards and hygienic bee stock provide good levels of control for the amount of effort and cost required to implement them, there is no chance of honey or wax contamination, MIPM provides an option to beekeepers who wish to produce honey organically, and it provides a means of control that reduces the probability

that resistance will develop. Organic honey would fill a market niche, and consumers would likely pay a premium for honey produced organically that would offset the increased cost of production.

Future experiments should test other control combinations such as modified bottom boards, hygienic bee stock, and a rotation of thymol three times in two years with Apistan one fall. Other options include using formic acid in place of thymol since formic acid is registered for use, formulations are more advanced, and overall cost is lower. Drone brood trapping also would be an interesting contribution to an IPM regime, although labor-intensive. A more in depth cost-benefit analysis of various systems would be beneficial to determine how useful these different methods are when used alone, and combinations could be selected based on those criteria.

The results of this experiment suggest that the emergency registration of coumaphos in the United States may not be justified. Resistance to coumaphos has been documented (Milani 1999, Elzen et al. 2000), and regulatory agencies today are attempting to phase out organophosphate compounds for pest management. Many agricultural commodities are moving away from single chemical reliance to either a rotational chemical IPM regime or, in some cases a mixed IPM regime. With the abundance of cultural, genetic, and chemical control methods available, and the rapid development of resistance to single chemical controls, movement into an IPM treatment regime for varroa control appears to be a viable and necessary option.

Chapter IV

Summary

Infestation of honey bee colonies by the parasitic mites *Varroa destructor* and *Acarapis woodi* has resulted in a world-wide decline in honey bee colonies, honey production, and crop pollination. Through the application of pesticides, especially Apistan for varroa control and menthol and vegetable oil for tracheal mites, these mites have been kept at sublethal levels. However, the development of resistance to chemical controls and the chronic infestation of bees by tracheal mites indicates that new methods of mite suppression need to be examined.

Botanical compounds such as thymol and blends of botanical compounds such as Magic 3 can suppress both tracheal and varroa mite populations. However, these compounds have a narrower efficacy window than Apistan, so the dose required to kill mites is closer to the dose that kills bees for botanicals than it is for Apistan. Botanicals also have more temperature-dependent release rates, and even with the best release devices mite control rarely is higher than 70%.

Farmers today increasingly rely on Integrated Pest Management, a technique that has potential to be effective in the management of honey bee pests as well. There are several non-chemical mite control methods available, but each alone does not provide levels of control that would sustain colony vitality. A combination of non-chemical controls along with chemical compounds may provide adequate mite control while slowing the development of resistance.

The experiments discussed in this paper address the development of an effective release device and the incorporation of such a device in an IPM treatment regime. Unacceptable levels of brood death and a decrease in overall adult bee and brood population were found with the Magic 3 sleeve, but thymol florist blocks provided approximately 35% control with no significant effect on adult bee or brood populations, and thus were selected for further experimentation.

I tested a mixed IPM treatment regime consisting of thymol florist blocks, modified bottom boards, and hygienic worker stock selection. This regime was compared to colonies treated with either thymol, Apistan, or a combination of the two (rotational IPM) administered in the spring and fall. Colony health was measured by examining adult bee populations, brood population, varroa levels, honey production, and colony survival over the 21 month treatment period. In measurements of all variables, there was no significant difference between treatments, with the sole exception being post-treatment mite levels in the fall of 2000.

This experiment demonstrated that honey bee colonies can survive with mite populations, as long as the mite population does not grow beyond survival threshold. It also demonstrated that an MIPM treatment regime can keep mite levels below the survival threshold for a length of time that would eliminate the need for annual administration of any single chemical. Apistan could be used in rotation with other chemical controls as part of an MIPM regime to flush mites and knock them back to near zero levels in the fall, but spring treatments can be varied. With fewer applications, resistance development may be slowed and reliance on single chemical use reduced.

Parasitic mites have hobbled the beekeeping industry. Honey prices and beekeeper morale are low. The number of beekeepers in North America has been dropping steadily since the mid-nineties, hobby beekeeping is becoming a rarity

(Agriculture and Agri-Food Canada, 2000) as the cost and complexity of treating for the varroa mite has put beekeeping out of the reach of many potential hobbyists, and the commercial industry is feeling a similar impact. Integrated pest management has been used successfully with dramatic results in countless agricultural applications. Farmers have become aware that reliance on a single control method for any pest is undesirable. Varroa lends itself well to IPM as a means to control the industry's number one pest, and IPM may prove to be what the industry needs to solve its current economic crisis.

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